

SPECIAL RELATIVITY

1 The two fundamental postulates

1. Physics is the same everywhere, and everything takes place in an intertial frame of reference.
2. The speed of light, c , is *always* $299\,792\,458\,\text{m s}^{-1}$, and is invariant. It does not change depending upon the observer's speed.

This means that for a train moving at $20\,\text{m s}^{-1}$ with a person on it shining an electric torch, to a stationary observer, the speed of light would not be $299\,792\,478\,\text{m s}^{-1}$. This conflicts with classical mechanics (CM), because in CM, velocities just add up.

2 Impacts of this

If c is to stay the same everywhere, then for the equation

$$c = \frac{\ell}{t} \quad (1)$$

wherein ℓ is the distance and t is the time taken to stay true, either distance or time must change. This is wherein length contraction and time dilation come in:

1. Length contraction: For an object with rest-frame length L_0 in motion, it shall shrink along the axis of motion. The measured length is denoted by L :

$$L = \frac{L_0}{\gamma} = L_0 \gamma^{-1} = L_0 \sqrt{1 - \frac{v^2}{c^2}} \quad (2)$$

So for an object with rest-frame length $1\,\text{km}$ moving at $269\,813\,212.2\,\text{m s}^{-1}$ ($0.9c$), the measured length would be:

$$L = 1 \cdot 10^3 \text{ m} \sqrt{1 - \frac{(269\,813\,212.2\,\text{m s}^{-1})^2}{(299\,792\,458\,\text{m s}^{-1})^2}} \approx 435.89 \text{ m} \quad (3)$$

2. Time dilation: Time is not a constant. It changes relative to an object's velocity,¹ and can be defined by the following equation, wherein Δt denotes the change in time:

$$\Delta t = \gamma \Delta t_0 = \quad (4)$$

¹This is a major theme of the 2014 film *Interstellar*.